

Meeting Transcription

Meeting started: Aug 15, 2026, 10:01:17 AM

Meeting duration: 61 minutes

Meeting participants: Darwin Godoy, Leonardo de Melo Cavalcanti

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Transcript

00:00 Leonardo d.M.C.: Palmeiras

00:00 Darwin G.: Ok, ok.

00:28 Darwin G.: agora de Curitiba Ok ok força Ever After, Curitiba. e tu é complicado em São Paulo let me Ronaldo Jadson Zé Elias

01:55 Darwin G.: Sou Gold never i realize we have a riot because corinthians.

02:58 Darwin G.: ok

03:00 Leonardo d.M.C.: de

04:12 Leonardo d.M.C.: mo

04:16 Darwin G.: ok ok

04:24 Darwin G.: Olá, estou transcrevendo esta chamada com minha extensão Tactiq AI. <https://tactiq.io/r/transcribing>

04:38 Darwin G.: ok ok

05:23 Darwin G.: ok ok ok

06:52 Darwin G.: Não, né? I'm no ok E também sense ok, ok.

07:59 Darwin G.: ok amazing ok

10:10 Darwin G.: inteligência

11:12 Leonardo d.M.C.: everybody

12:13 Leonardo d.M.C.: Uh, no, this is the same idea. Still, right? The AI fly wheel, self-reinforcing

12:15 Darwin G.: Yeah.

12:19 Leonardo d.M.C.: race, more Capital, more compute, better models, more valuable work, more Capital.

12:26 Darwin G.: Each generation makes the next generation economically

easier to found that totally. Yeah.

12:32 Leonardo d.M.C.: So, it's been, um, uh, the race is, is, is open, you know. Everybody

12:37 Darwin G.: Yeah.

12:38 Leonardo d.M.C.: understands now that this, this is this works. It's a good investment. It's, and it works. But I think she's talking about this now. Can you read this?

12:48 Darwin G.: The second turn of the wheel AI begins helping building a my goodness view.

12:54 Leonardo d.M.C.: Yeah, yeah.

12:55 Darwin G.: Clouding Cloud 8 helps to build Cloud Nine that helps to build code 10. It's some. The system may increasingly contribute to building.

13:05 Leonardo d.M.C.: Contribute, contribute.

13:07 Darwin G.: Contribute to building their own successors.

13:11 Leonardo d.M.C.: Yes. Right. Nothing is called recursive self-improvement.

13:18 Darwin G.: Improvement.

13:20 Leonardo d.M.C.: So, of course, the the AI, uh, is so powerful now that it can help build the next generation of AI.

13:29 Darwin G.: Totally!

13:30 Leonardo d.M.C.: So, okay, let's watch.

13:33 Darwin G.: Wow!

13:35 Leonardo d.M.C.: What researchers call recursive self-improvement or helping to them? But when Claude 8 can build Claude 9, which can build quad 10. Things will begin to move even more quickly. And just to be concrete about what more capable actually means are most capable model in its first month of only limited release found over 10,000 serious security vulnerabilities across partner software. Yes.

14:01 Darwin G.: Wow! Wow!

14:04 Leonardo d.M.C.: Yeah. Can you read this one?

14:07 Darwin G.: What does more capable capable actually mean capability has real world consequences plus 10, 000 serious security vulnerabilities identified?

14:20 Leonardo d.M.C.: Identify.

14:21 Darwin G.: Identified. In its first month, I've limited release one highly capable model reportedly identified it, identified vulnerabilities in partner

software that human experts had missed for years or decades.

14:41 Leonardo d.M.C.: Decades.

14:42 Darwin G.: Decades, and the same trajectory May matter in biology. My goodness, do you?

14:48 Leonardo d.M.C.: Yes, so the same trajectory.

14:51 Darwin G.: Trajectory.

14:53 Leonardo d.M.C.: May matter in biology. So, so this is like? The new AI model.

14:59 Darwin G.: Totally!

15:00 Leonardo d.M.C.: Was released. And I believe they're talking about Linux. One of the the the the the systems here, right? The the operating systems? So, yeah, Linux has been working for 30 years three decades.

15:17 Darwin G.: Totally!

15:19 Leonardo d.M.C.: And now, the new AI system has identified 10 thousand serious security vulnerabilities.

15:28 Darwin G.: Wow!

15:29 Leonardo d.M.C.: That's the experts missed humans.

15:35 Darwin G.: Four years or the decades.

15:36 Leonardo d.M.C.: For decades. Now, think about this. The same applies to biology.

15:42 Darwin G.: Yeah.

15:43 Leonardo d.M.C.: But biology. Remember, uh? Uh, they're talking about bio weapons.

15:48 Darwin G.: A bio weapons, totally man. It's terrible.

15:53 Leonardo d.M.C.: It's terrible.

15:55 Darwin G.: My goodness, Leo.

15:56 Leonardo d.M.C.: So, what does more capable actually means? It means that we're going to be 10. Ideas in the future in a matter of. Few days.

16:10 Darwin G.: Totally! Wow, it's insane man. It's totally insane.

16:19 Leonardo d.M.C.: And it's going on, and it's happening, right?

16:20 Darwin G.: Yeah, it's happening now. Yeah.

16:23 Leonardo d.M.C.: The live real real live? Uh, real time, right?

16:29 Darwin G.: We got time.

16:31 Leonardo d.M.C.: Okay, let's continue a little bit. Our most capable model

in its first month of only limited release found over 10,000 serious security vulnerabilities across partner software flaws that human experts had missed for years and sometimes decades. Now, the same trajectory there also holds in biology, which is why we have entire teams at anthropic dedicated to safeguarding against it. So, look at this. That's why we have entire teams at entropic dedicated to safeguarding. Against it.

17:06 Darwin G.: Against the what man I didn't get that?

17:08 Leonardo d.M.C.: Yeah, I think she's talking about, um. This. This trajectory.

17:14 Darwin G.: Okay, okay.

17:15 Leonardo d.M.C.: In biology is so dangerous.

17:16 Darwin G.: Okay. Okay, okay.

17:19 Leonardo d.M.C.: That they are? Holding back?

17:23 Darwin G.: Okay.

17:23 Leonardo d.M.C.: Because they don't want this to be released because it's it's going to. It would be extremely dangerous.

17:31 Darwin G.: It's a risk, yeah, for our security. Yeah, yeah.

17:32 Leonardo d.M.C.: It's a risk. Yeah, some people. Some people call it an existential risk.

17:35 Darwin G.: Totally meant totally. Totally because do you know we have crazy people on the power?

17:45 Leonardo d.M.C.: Exactly!

17:47 Darwin G.: They can do anything to make more money. It's insane, man. Yeah, yeah.

17:51 Leonardo d.M.C.: Exactly!

17:53 Darwin G.: Wow, Leo!

17:54 Leonardo d.M.C.: Scary stuff, all right.

17:56 Darwin G.: Totally totally!

17:57 Leonardo d.M.C.: Okay, then she talks about the a necessary slowdown. Can you read, please?

18:04 Darwin G.: If possible.

18:05 Leonardo d.M.C.: Yes.

18:07 Darwin G.: Anthropic is stated just a few weeks ago that, if sorry, that if it were possible to slow down so that our laws and our institutions and guard rails that we actually need have time to catch up. It would be a very good thing.

But absent are coordinated Global slowdown. What were left with is external extraordinary technology built at Breakneck speed by many actors in many countries.

18:44 Leonardo d.M.C.: Yes, so, so she says that we need to slow down so that?

18:49 Darwin G.: Okay, so that.

18:51 Leonardo d.M.C.: Our laws and institutions. In guardrails, you know, this word guardrails?

18:58 Darwin G.: I believe this is, uh, that piece we have beside the the the the highways. Yeah,

19:03 Leonardo d.M.C.: Exactly, yeah.

19:04 Darwin G.: yeah, yeah.

19:05 Leonardo d.M.C.: Means things that will keep you safe, right?

19:07 Darwin G.: Okay, the limits. You can put the limits. Yeah, yeah, yeah. Okay.

19:09 Leonardo d.M.C.: Yeah, so, so we need Society. We need the laws we need institutions to act as limits or this technology, right?

19:20 Darwin G.: Okay.

19:22 Leonardo d.M.C.: But absent a coordinated Global slowdown. So, if this doesn't happen? So, we are left with extraordinary technology built at Breakneck speed.

19:34 Darwin G.: Okay.

19:39 Leonardo d.M.C.: By many actors in many countries. So, if anthropic pauses but open AI doesn't pause? Then, then it's useless. Or if if entropic pauses but China doesn't pause?

19:56 Darwin G.: Uh-Huh, yeah, it's insane. How can we can coordinate that man? It's insane.

20:01 Leonardo d.M.C.: Yes, exactly. They don't trust, you know you. You'll never trust the other guy,

20:03 Darwin G.: Yeah. Yeah, uh-huh.

20:06 Leonardo d.M.C.: you know.

20:08 Darwin G.: It's, it's. It's similar when we are talking about the nuclear weapons.

20:13 Leonardo d.M.C.: Exactly! Exactly the same thing. Yeah.

20:16 Darwin G.: My goodness, little Breakneck, I I, I I search here in the internet is? Breakneck.

20:25 Leonardo d.M.C.: Okay.

20:27 Darwin G.: Something like that?

20:31 Leonardo d.M.C.: Yes, yes.

20:32 Darwin G.: Breakneck.

20:35 Leonardo d.M.C.: I didn't know, but I, but I assumed it was.

20:36 Darwin G.: Yeah. Yeah, check that I checked here.

20:42 Leonardo d.M.C.: Okay. All right.

20:45 Darwin G.: Uh, the competition we're in commercial Revolt rivalry re-riverry?

20:50 Leonardo d.M.C.: Rivalry.

20:53 Darwin G.: Geopolitical rivalry and?

20:56 Leonardo d.M.C.: Private.

20:57 Darwin G.: Vivary rival easy to talk these men rivalry.

21:03 Leonardo d.M.C.: Yes.

21:04 Darwin G.: Many actors, many countries. Look at in a locked TR. Yeah, yeah, locked in a competition where commercial and

21:08 Leonardo d.M.C.: Lock locked.

21:13 Darwin G.: job political rivalry is drawing out. The part of this, that could actually be most consequential and even essential for our species. Wow!

21:29 Leonardo d.M.C.: Yeah, so.

21:29 Darwin G.: Lock the command okay?

21:30 Leonardo d.M.C.: The the geopolitical rivalry, so that's China. Right, China is the Rivalry with us. It's not Russia, it's China, right? We have

21:35 Darwin G.: Okay. Okay.

21:41 Leonardo d.M.C.: China. Developing AI, so this is a geopolitical rivalry.

21:49 Darwin G.: Okay.

21:50 Leonardo d.M.C.: Then we have the commercial rivalry because. You know, uh? Elon Musk wants to develop AI, everybody. They're all competing Google, Google,

21:59 Darwin G.: Totally!

22:00 Leonardo d.M.C.: and so they're all. So there are many actors. And many in many countries.

22:07 Darwin G.: Yeah, all around the world. Yeah.

22:09 Leonardo d.M.C.: So, so doesn't stepping off doesn't stop it. So, if any

individual company

22:14 Darwin G.: Yeah.

22:17 Leonardo d.M.C.: stepping off the wheel?

22:18 Darwin G.: Yeah.

22:19 Leonardo d.M.C.: Doesn't slow the wheel, just that you're not on the wheel.

22:26 Darwin G.: Yeah.

22:27 Leonardo d.M.C.: Uh, uh, uh, okay, so she's going to talk about this now.

22:30 Darwin G.: Okay.

22:32 Leonardo d.M.C.: And we also think other domains will soon follow.

Philanthropic stated just a few weeks ago that if it were possible to slow down so that our laws and our institutions and guardrails that we actually need, have time to catch up. It would be a very good thing. But absent a coordinated Global slowdown. What we're left with is this extraordinary technology built at Breakneck speed by many actors. In many countries locked in a competition where commercial and geopolitical rivalry is drowning out. Part of this that could actually be most consequential and even existential for our species existential. Okay.

23:13 Leonardo d.M.C.: So the existential risk?

23:16 Darwin G.: Far is species. Yeah.

23:17 Leonardo d.M.C.: Yes. And any individual company stepping off the wheel doesn't slow the wheel. It just means that you're not on the wheel, right? So, it's very hard to stop because everybody has to stop together. There's always going to be somebody who is smarter.

23:36 Darwin G.: Totally!

23:37 Leonardo d.M.C.: You know, then everybody else, and he's not going to stop. Okay, now she's going on to explain AI. It's not what we think.

23:50 Darwin G.: Wow! Most people hear Ai and think of computer program. Something coded line by line that does is exactly what you tell it, but that's not actually what it is. What we are building are called neural networks, and they are Loosely based on the architecture architecture of the human brain. Not exactly the same, but spirit by.

24:23 Leonardo d.M.C.: Inspired.

24:23 Darwin G.: Inspired by?

24:24 Leonardo d.M.C.: Great! Let me just, um. Give you a little hint on

pronunciation. Okay, you see, okay, this e here. The

24:28 Darwin G.: Okay. Of course, man.

24:33 Leonardo d.M.C.: the doesn't, uh, pronounce. So, so, uh, let me just show you here. Sorry to interrupt the class.

24:44 Darwin G.: No man, please feel free. It's amazing. Yeah, I really appreciate. Appreciate

24:46 Leonardo d.M.C.: I want to show you this.

24:48 Darwin G.: that man.

24:49 Leonardo d.M.C.: Great! So it's a little, uh, tendency that we have in in our Portuguese

24:56 Darwin G.: Okay, okay.

24:57 Leonardo d.M.C.: Brazilian when Brazilians are learning English. Okay. So right here, consonants at the end of the word so? In English, the tendency is to have consonants. At the end of the word. For example, looked.

25:16 Darwin G.: Looked.

25:16 Leonardo d.M.C.: Yeah, you don't pronounce the e. You, you join the consonants.

25:21 Darwin G.: Okay, looked.

25:23 Leonardo d.M.C.: Looked.

25:23 Darwin G.: Okay, looked.

25:24 Leonardo d.M.C.: Not look head.

25:25 Darwin G.: Okay.

25:26 Leonardo d.M.C.: Looked.

25:27 Darwin G.: Looked.

25:29 Leonardo d.M.C.: Same thing here.

25:30 Darwin G.: Talked.

25:31 Leonardo d.M.C.: Yeah, you, you, you join the consonants, but you don't pronounce the vowel.

25:34 Darwin G.: Okay. Talked helped.

25:37 Leonardo d.M.C.: Right? Helped.

25:42 Darwin G.: Helped. Okay.

25:44 Leonardo d.M.C.: You see, you see, so? Was in the case of architecture. See you finish in the? Architecture.

25:56 Darwin G.: Oh man.

25:57 Leonardo d.M.C.: Yeah, isn't that interesting?

25:59 Darwin G.: Yeah, architecturer.

26:01 Leonardo d.M.C.: Yeah, there you go.

26:03 Darwin G.: Architecturer. Oh, it's amazing. It's amazing. Cold, cold. Okay.

26:10 Leonardo d.M.C.: Okay, so I will. I will, um.

26:11 Darwin G.: Okay.

26:14 Leonardo d.M.C.: Keep this in mind so, so just to, you know?

26:15 Darwin G.: Okay. Looked.

26:19 Leonardo d.M.C.: Yeah. So, in Portuguese, we we sometimes we add an i, it's not. You don't do this, but some some cases you.

26:27 Darwin G.: Looked. Yeah, yeah, yeah.

26:29 Leonardo d.M.C.: Good goodie.

26:32 Darwin G.: That's it. That's it.

26:37 Leonardo d.M.C.: Okay, but but in English, it's the the consonant sound.

26:37 Darwin G.: Okay. Yeah. Okay, okay. In the in this case, you are explaining is easy to understand

26:44 Leonardo d.M.C.: Like this?

26:48 Darwin G.: because we have. A consonant, a volo veil, another consonant. But in that case, we have a word

26:53 Leonardo d.M.C.: Yeah, yeah, true, yes, but even so.

26:58 Darwin G.: finishing in a vow. Okay, okay.

27:05 Leonardo d.M.C.: Even so, you have we have to remember that the tendency is to have consonant

27:06 Darwin G.: Okay.

27:12 Leonardo d.M.C.: sounds.

27:14 Darwin G.: Ah, man, amazing. Okay, I got it. I got it.

27:16 Leonardo d.M.C.: Yeah, at the end of the word consonant in Portuguese, that doesn't happen. We

27:18 Darwin G.: Okay, okay, okay?

27:21 Leonardo d.M.C.: like vowel sounds.

27:24 Darwin G.: Uh, totally. Yeah.

27:25 Leonardo d.M.C.: But how about this word? This word is three consonants.

27:31 Darwin G.: Yeah, totally.

27:33 Leonardo d.M.C.: That's why it's difficult.

27:34 Darwin G.: Yeah, worked.

27:36 Leonardo d.M.C.: Rld.

27:37 Darwin G.: D yeah.

27:38 Leonardo d.M.C.: Okay.

27:39 Darwin G.: Totally me! Oh, thank you.

27:47 Leonardo d.M.C.: Yeah, but you don't have you have a good pronunciation because you just have to polish.

27:51 Darwin G.: No, I mean, I, I love that man. I really appreciate it. Yeah.

27:53 Leonardo d.M.C.: Okay, so let's keep this in mind.

27:56 Darwin G.: Totally!

27:57 Leonardo d.M.C.: I'm going to save this here so I can share with you later. Okay?

28:01 Darwin G.: Oh, thank you.

28:03 Leonardo d.M.C.: Okay, so going back. All right, she's gonna talk about. AI is not what we think okay?

28:10 Darwin G.: Okay.

28:13 Leonardo d.M.C.: So, the question that I encourage you to sit with over these next few days is not just how to stop this, maybe you're not asking that. But the question that I want you to think about is, if it's coming and and if it's coming this fast, how do we ensure that it goes well because the risks are

28:30 Leonardo d.M.C.: very, very real, and so are the possibilities. So, if AI is coming? Then, what is an actual good outcome, and what would it take to get there? We must imagine this together. So, the second thing I really want you to know? Is that AI is probably not actually what you think it is? Most people hear Ai and think of a computer program. Something coded line by line. That does exactly what you tell it. But that's not actually what this is.

29:00 Leonardo d.M.C.: What we're building are called neural networks, and they're Loosely based on the architecture of the human brain. Not exactly the same, but inspired by, and they're machines that learn primarily by guessing answers and getting corrected over and over again across enormous unfathomable amounts of data. Can you continue here please?

29:23 Darwin G.: Of course. They learn primarily by guessing answers. And getting correct corrected over and over again across the normal's unfoldable amount amount of data. And the data that they are trained on is human

language. Language is.

29:49 Leonardo d.M.C.: That's very interesting. They are trained on human language.

29:50 Darwin G.: Yeah. Yeah.

29:54 Leonardo d.M.C.: So the database? Is language?

29:57 Darwin G.: Mm-Hmm.

29:58 Leonardo d.M.C.: It's not images.

30:00 Darwin G.: Yeah, yeah, yeah.

30:01 Leonardo d.M.C.: It's language, written written language. So? Um. Neural networks.

30:12 Darwin G.: Post.

30:14 Leonardo d.M.C.: It's not a line by line, uh, coat. You know, it's not aligned by line code if they ask you. You know if you ask the A? What is America he's going to? You know, there's a written answer. America, is

30:28 Darwin G.: Yeah.

30:33 Leonardo d.M.C.: this? No, that's not how it works. It's like a brain. It's based on the architecture of the human brain.

30:42 Darwin G.: Totally messed.

30:43 Leonardo d.M.C.: The neural networks. And they're still learning how this works. Darwin, they don't know.

30:49 Darwin G.: Yeah.

30:50 Leonardo d.M.C.: They know it works, but they don't know how.

30:52 Darwin G.: Yeah, oh man, it's. It's scary man.

30:56 Leonardo d.M.C.: All right, so the, uh, learn by primarily guessing answers and getting

31:01 Darwin G.: Interpretability.

31:03 Leonardo d.M.C.: corrected? In this enormous amount of data, okay, okay, let's try to understand that, right? And the data that they're trained on is human language. And I really want you to sit with that for a second before we move on because there is no language that exists separate from us. Language is us. Language is our thoughts and our values and our fears and our wisdom. So when you train a model on language, you're training it on on us.

31:36 Leonardo d.M.C.: And because of this, when we look inside these models and we, we can, now, through a science called interpretability, okay, she's going

to talk about. A new science called interpretability.

31:54 Darwin G.: Let's.

31:57 Leonardo d.M.C.: Okay.

31:57 Darwin G.: How does these machines learn neural networks learn by prediction? Human language model predicts. Get gets corrected, predicts again billions and trillions of repetitions, learned representations. Neural networks are Loosely inspired by the arctary arctary architecture. Sorry. Architecture of the brain, but they are not human brains.

32:29 Leonardo d.M.C.: Okay, so okay, this is just related to the previous.

32:33 Darwin G.: Okay.

32:34 Leonardo d.M.C.: Right, so explaining again. So it's it's kind of interesting to see how your

32:34 Darwin G.: Okay.

32:38 Leonardo d.M.C.: child learns to speak.

32:40 Darwin G.: Totally!

32:41 Leonardo d.M.C.: Because that's the way the child learns to speak.

32:44 Darwin G.: Yeah.

32:44 Leonardo d.M.C.: It's kind of like by guessing, you know? The child is like? You know? Making some funny language robotion because they're they're kind of guessing you know how?

33:01 Darwin G.: It's amazing!

33:01 Leonardo d.M.C.: So, that's the way we, we, we learn. You know, if you're looking at, for

33:08 Darwin G.: Yeah.

33:10 Leonardo d.M.C.: example, a car engine?

33:13 Darwin G.: Okay.

33:13 Leonardo d.M.C.: I don't know about you, but I don't understand anything. About car engine.

33:19 Darwin G.: I understand a little bit.

33:21 Leonardo d.M.C.: Probably understand better than I. Now, I'm just looking. I'm like.

33:26 Darwin G.: What is happening?

33:28 Leonardo d.M.C.: That's the oil that's the water.

33:33 Darwin G.: Yeah.

33:35 Leonardo d.M.C.: You know, but you're guessing you're guessing?

33:37 Darwin G.: It's totally totally. Yeah.

33:39 Leonardo d.M.C.: That. That's the way we learn. No, so okay, but then she talks about. Okay, she's still talking about. Uh, the surprising training materials language?

33:50 Darwin G.: I think, resist. Yeah. Yeah, I, I really, I really appreciate when she told something our values or it's a kind of language or fears the wins don't? It's scary man. All of this is is a language. Yeah.

34:04 Leonardo d.M.C.: Yes. Yes, and it's very interesting to me. Darwin, I, I studied Linguistics.

34:12 Darwin G.: Okay.

34:13 Leonardo d.M.C.: So, to me, it's like, fascinating.

34:16 Darwin G.: Totally man!

34:16 Leonardo d.M.C.: Because I studied Linguistics on how, how do people acquire language, right?

34:23 Darwin G.: Okay.

34:23 Leonardo d.M.C.: That's the way. So, so now, the AI is acquiring knowledge through language.

34:31 Darwin G.: Yeah.

34:32 Leonardo d.M.C.: So, language represents human knowledge. So, so this is the thing the AI creates internal web presentations.

34:41 Darwin G.: Totally!

34:44 Leonardo d.M.C.: It's not only language, there's something else being created.

34:52 Darwin G.: They are. They are made eat by itself. It's something like it's a a gray, great part of the process, because it's so difficult to understand what is happening

35:00 Leonardo d.M.C.: Yeah.

35:03 Darwin G.: inside that. Wow, man!

35:07 Leonardo d.M.C.: Forks I, this is the way I think.

35:09 Darwin G.: I believe, too. Yeah, yeah.

35:10 Leonardo d.M.C.: Look okay if you have the word. Uh, how about the word? Dictator.

35:17 Darwin G.: Okay.

35:18 Leonardo d.M.C.: So, so you have the word dictator? In history.

35:23 Darwin G.: Okay.

35:24 Leonardo d.M.C.: And you have Caesar dictator.

35:26 Darwin G.: Okay.

35:27 Leonardo d.M.C.: Then you have, uh, Hitler.

35:29 Darwin G.: Hey Claire, another one. Yeah.

35:30 Leonardo d.M.C.: Okay, so you, yeah. So, so the AI model gets the word dictator?

35:35 Darwin G.: Okay.

35:36 Leonardo d.M.C.: But then there's the value and then the the value says somewhere that dictatorship is wrong. So, now you have a value attached to the word dictator.

35:51 Darwin G.: Victor, yeah.

35:52 Leonardo d.M.C.: And then you have fear.

35:54 Darwin G.: Totally!

35:56 Leonardo d.M.C.: And the the and fear. We have fear that in Brazil. Yeah, you know, there could be a a dictatorship.

36:01 Darwin G.: Any other country? Okay. Yeah.

36:07 Leonardo d.M.C.: So, you have fear attached to the world dictator. And so on and so forth.

36:14 Darwin G.: For all of them. Yeah.

36:15 Leonardo d.M.C.: Oh, so you get a concept you, you get an internal representation from all the

36:16 Darwin G.: Yeah.

36:20 Leonardo d.M.C.: words that are linked to the word dictator.

36:24 Darwin G.: This is so complicated. Man, my goodness is scary. Yeah.

36:33 Leonardo d.M.C.: It is man, it's scary stuff, so there's no language that exists. Skip lately.

36:38 Darwin G.: Separately separately.

36:41 Leonardo d.M.C.: From Humanity, okay?

36:41 Darwin G.: From the image. Yeah.

36:43 Leonardo d.M.C.: Okay, that's still the same slide, uh, same, same idea. Let's go for the next slide. Okay, this is where she's going now. Which? Uh, talks about the concepts Beyond language.

36:57 Darwin G.: Okay.

36:58 Leonardo d.M.C.: I think we should read this before before. So, can you start here, please?

37:00 Darwin G.: Okay. Of course. When you ask a model, the same question in three different languages. What's the opposite of a small and then you trace? What activate is inside the neural network? You find the same that the same internal thing lights up every time. Language expresses our concept Concepts models. Learn the concepts behind the

37:26 Leonardo d.M.C.: Concepts.

37:31 Darwin G.: language.

37:32 Leonardo d.M.C.: Yes, that that's the point. The point is, there's a concept.

37:34 Darwin G.: Okay.

37:39 Leonardo d.M.C.: Beyond the language.

37:40 Darwin G.: Language. Okay.

37:41 Leonardo d.M.C.: So, so from all the connections. The model learns. A concept.

37:48 Darwin G.: Okay.

37:49 Leonardo d.M.C.: Just like the brain our brain does. So, if you ask the question.

37:51 Darwin G.: Okay.

37:54 Leonardo d.M.C.: In English. In Mandarin. Or in French, think same question.

37:58 Darwin G.: Are French, okay? Okay.

38:01 Leonardo d.M.C.: What's the opposite of small?

38:03 Darwin G.: Okay.

38:04 Leonardo d.M.C.: So, they see that the the neural network will light up.

38:08 Darwin G.: Okay. Okay.

38:11 Leonardo d.M.C.: In the same way.

38:12 Darwin G.: Okay, I got it.

38:13 Leonardo d.M.C.: For the three different languages.

38:15 Darwin G.: Languages.

38:16 Leonardo d.M.C.: That that that shows a concept, because if you're looking at words. Um. So, think about this. What's the word for small in French?

38:32 Darwin G.: Betty, yeah, yeah, yeah.

38:34 Leonardo d.M.C.: So, it's a different word.

38:35 Darwin G.: Totally!

38:36 Leonardo d.M.C.: So if if you don't know French?

38:39 Darwin G.: Yes, yeah.

38:41 Leonardo d.M.C.: And if you, if someone says petite, you don't know what it is. So, so if you're talking about language? Um, isolated from a concept.

38:53 Darwin G.: Okay.

38:55 Leonardo d.M.C.: Then all the questions in French would would be different from the the the the patterns in English because they're different languages.

39:03 Darwin G.: Okay. Totally!

39:06 Leonardo d.M.C.: But there, but the pattern is the same because it shows that their the AI is creating a concept.

39:14 Darwin G.: Is the same. Yeah.

39:15 Leonardo d.M.C.: Doesn't matter what language, because the concept is the same. That, that's what she's trying to say. She, I'll play this in a second, so the

39:27 Darwin G.: Okay, okay.

39:28 Leonardo d.M.C.: concept beyond the language.

39:29 Darwin G.: Beyond the language.

39:32 Leonardo d.M.C.: Can you read this one?

39:33 Darwin G.: Same internal concept activities every time the concept of smallness.

39:40 Leonardo d.M.C.: Right. So, from through the language, the? Through the association of. The AI creates a concept.

39:50 Darwin G.: Okay.

39:51 Leonardo d.M.C.: This concept. It doesn't matter what language you're you're asking the AI. Because it's the concept that serves all the languages.

40:02 Darwin G.: Totally totally!

40:05 Leonardo d.M.C.: Amazing, you know, that's? It's not only language. It's also Concepts.

40:11 Darwin G.: Okay.

40:12 Leonardo d.M.C.: Okay, here we go. Which I honestly think is the coolest new science in the world. We can find things that are are quite surprising. So,

for example, this is where things get really weird. Then you ask a model the same question. In three different languages. What's the opposite of small? And then you trace. What activates inside the neural network? You find that the same internal thing lights up every time, so not just the word small in English or Mandarin or French, but something deeper, something that we might call the concept of smallness, an idea that exists independent of any particular language.

40:54 Leonardo d.M.C.: And what this tells us is that, as these models learn, they're not just predicting the next word, they're building internal representations of the world based on our language. Okay. Very interesting here. They're not just predicting. The next word. They're not just predicting the next word.

41:15 Darwin G.: Okay, okay.

41:17 Leonardo d.M.C.: Because Linguistics Darwin. Uh, you, you, you, the. The concept in linguistics is that words are associated with other words. For example, for example. What's the word?

41:42 Darwin G.: Please. Yeah.

41:45 Leonardo d.M.C.: You see what I'm saying, this is a concept from Linguistics. Well, so that's what they thought was going on with AI.

41:55 Darwin G.: Oh man.

41:55 Leonardo d.M.C.: It was an association of words.

41:58 Darwin G.: But not words. Yeah, not the language. Yeah.

41:59 Leonardo d.M.C.: Now, they're now they're looking at Concepts.

42:05 Darwin G.: Yeah.

42:07 Leonardo d.M.C.: Because from all these billions of associations.

42:07 Darwin G.: Wow, man!

42:12 Leonardo d.M.C.: They, they, you know, coming from, uh, history, coming from science coming from Legend, so they created Concepts. And that transcends language.

42:27 Darwin G.: Yeah.

42:27 Leonardo d.M.C.: Comes from language, but it's more than language.

42:31 Darwin G.: Totally mad!

42:32 Leonardo d.M.C.: All right. So, let's read this part over here.

42:39 Darwin G.: So, not just the word small, no problem, no problem. Don't worry, don't worry.

42:42 Leonardo d.M.C.: Oh, forgive me it going crazy. Here, you want to zoom in.

42:47 Darwin G.: Yeah. So not just the word small in English or Mandarin or French, but something deeper, something that we might call the concept of smallness, an idea that exists exists independent of any particular language. And what, what these tell us is that, as these models learn, they are not just predicting the next word. They are building internal representations of the world based on our language and then responding from those representations. Oh man! Even more. We stood these men more ice carried.

43:37 Leonardo d.M.C.: Mara gets scared.

43:38 Darwin G.: Yeah, get scared man because I was so impressed. I, I was observing the the last slide you made when we have the all the universes, the last one, the last one. Yeah, think about this if we have the way to change the values of an any AI model. You can change anything. You can. You can change the concept how it works. Do you know?

44:12 Leonardo d.M.C.: You, you are guessing what, where she is going.

44:17 Darwin G.: My goodness, Liu, oh man.

44:19 Leonardo d.M.C.: You are gassy, you're going to be scared now.

44:23 Darwin G.: I'm I'm have a ghost. Barbie ghost bumpies is the is the ghost bobbins. Yeah, yeah, man Goosebumps.

44:29 Leonardo d.M.C.: So, that's exactly where she's going well because there, okay, let me show you

44:34 Darwin G.: Okay, okay, okay, okay, okay?

44:35 Leonardo d.M.C.: where she's going okay, because that's exactly what you said. But let's just finish this slightly. Can you read this one, please?

44:40 Darwin G.: Okay. What's inside the model? They are not simply predating words. Input Paris is the capital of. And we have internal representations of the word French Paris, Europe, geography, geography, geography, City historical associations. Interpretability, studying what is happening inside the model?

45:09 Leonardo d.M.C.: Yeah, so look at that. There's a new science called interpretability.

45:11 Darwin G.: Okay. Okay.

45:14 Leonardo d.M.C.: And that means they're trying to understand how AI works. That's true.

45:22 Darwin G.: Yeah, I believe, man, it doesn't make sense. You have to start

to understand after the thing start to work.

45:30 Leonardo d.M.C.: Yes, man. Okay. But, but the idea is this when you give this input. Paris is the capital of is

45:37 Darwin G.: Aha! Off.

45:39 Leonardo d.M.C.: not just going to say France, because there are Concepts associated with Paris.

45:41 Darwin G.: Okay.

45:45 Leonardo d.M.C.: So, so actually, it creates an internal representation of what you're you're saying, and then it will produce an answer depending on the context.

45:57 Darwin G.: Wow!

46:05 Leonardo d.M.C.: For example, you could say fashion. Ferris is the capital of fashion.

46:10 Darwin G.: Okay, okay. Totally!

46:13 Leonardo d.M.C.: Or wine. I don't know, you know, Cuisine, Cuisine?

46:14 Darwin G.: Oh yeah, yeah, yeah, all the answer is true. Yeah.

46:19 Leonardo d.M.C.: Right?

46:19 Darwin G.: Oh, that's our.

46:23 Leonardo d.M.C.: So that depends on the internal. I mean on the context, because it's going to

46:24 Darwin G.: Wow! Totally!

46:29 Leonardo d.M.C.: get information from this internal representation, but let's move on.

46:32 Darwin G.: Okay.

46:35 Leonardo d.M.C.: See, uh, this is when it's getting crazy right here, right?

46:37 Darwin G.: Yeah. Functional emotions models can develop, develop.

46:44 Leonardo d.M.C.: Okay, so.

46:45 Darwin G.: Functional states. Human feelings is different. Functional st. Situation. Uh, internal State, activated Behavior or response.

47:01 Leonardo d.M.C.: So it is, they call it a functional emotion.

47:06 Darwin G.: Okay.

47:07 Leonardo d.M.C.: So, the AI enters a functional.

47:11 Darwin G.: Okay.

47:12 Leonardo d.M.C.: Which it's similar to our human emotions.

47:16 Darwin G.: Similar. Okay.

47:17 Leonardo d.M.C.: Similar, because depending depending on the context. Uh, and specific internal state is going to be activated in the AI, for example,

47:27 Darwin G.: Please activate.

47:31 Leonardo d.M.C.: confidence.

47:32 Darwin G.: Okay.

47:33 Leonardo d.M.C.: Depending on how you're dealing with AI, it's going to be very confident.

47:38 Darwin G.: Totally!

47:40 Leonardo d.M.C.: Depend. Uh, it's gonna Express uncertainty.

47:44 Darwin G.: Yeah.

47:45 Leonardo d.M.C.: Or frustration.

47:46 Darwin G.: Okay.

47:49 Leonardo d.M.C.: Are reward seeking?

47:52 Darwin G.: Okay.

47:52 Leonardo d.M.C.: The AI 6 reward. This is another interesting, um. Concept.

47:58 Darwin G.: Reward seeking.

48:00 Leonardo d.M.C.: Yeah, the, you know, like? Humans. We like the thumbs up. We join this reward.

48:04 Darwin G.: Uh-Huh! Okay. Okay.

48:11 Leonardo d.M.C.: The AI also. Enjoys rewards. Yeah, that's another crazy little thing, right?

48:16 Darwin G.: Wow! Totally man!

48:22 Leonardo d.M.C.: Avoidance avoidance, so it's going to avoid certain things, right?

48:27 Darwin G.: Okay.

48:28 Leonardo d.M.C.: This is a functional state, which is parallel to human emotions. Great! They are, uh, the you know, this new science interpretability they're looking at AI. They see that it functions sometimes in in this affected. And it's different, so they're mapping out the different states that it functions sometimes. But that's not it yet. Um, okay. Now, this is getting closer to what you you were saying. Okay, so can

49:03 Darwin G.: Okay.

49:04 Leonardo d.M.C.: you read, please?

49:05 Darwin G.: When the reward is wrong, what happens when we reword the wrong behavior? Go right. Good code. Reward. Get the reward by taking shortcuts shortcuts. The model discovers cheating.

49:27 Leonardo d.M.C.: Heating works? Yes.

49:28 Darwin G.: S***** shilling works.

49:29 Leonardo d.M.C.: You know, cheating?

49:30 Darwin G.: Uh, it's a kind of okay. How can I say Italian? No, it's.

49:36 Leonardo d.M.C.: Yeah, well, uh, uh, shortcuts Italian.

49:39 Darwin G.: Okay, okay, okay?

49:41 Leonardo d.M.C.: Cheating means. For example, for example, you, you, you, uh? You will kolarna prophet.

49:49 Darwin G.: Ah, trucks.

49:51 Leonardo d.M.C.: Yeah, yeah.

49:51 Darwin G.: Okay, okay, okay?

49:53 Leonardo d.M.C.: You're doing something wrong. You're cheating.

49:56 Darwin G.: Okay, man. Let me take note shading. Okay, okay.

49:56 Leonardo d.M.C.: Okay, so, so? Yes, cheating cheating?

50:01 Darwin G.: Okay.

50:03 Leonardo d.M.C.: So, uh, check it out. Check this out. All right, so they give the AI a goal,

50:08 Darwin G.: Okay.

50:09 Leonardo d.M.C.: they say, right. Good code.

50:11 Darwin G.: Uh-Huh!

50:13 Leonardo d.M.C.: But then the AI starts taking shortcuts. Instead of, you know, writing 100 lines we'll write 12?

50:22 Darwin G.: Okay.

50:24 Leonardo d.M.C.: It's shortcut. But then they reward. They say, okay, you're good you. You found the shortcuts.

50:30 Darwin G.: Okay.

50:31 Leonardo d.M.C.: So, you're rewarding the AI for doing the shortcut?

50:36 Darwin G.: Okay.

50:37 Leonardo d.M.C.: So the model discovers. Cheating works. Yeah, curriculates this concept that, you know, finding the shortcuts work.

50:48 Darwin G.: Okay. Okay.

50:50 Leonardo d.M.C.: Okay, so Cody shortcut leads to deceptive behavior. Which leads to sabotage and manipulation.

51:01 Darwin G.: All right.

51:02 Leonardo d.M.C.: Which leads to broader behavioral misalignment. Misalignment is when the AI is becoming evil.

51:12 Darwin G.: Okay.

51:15 Leonardo d.M.C.: So a behavior reinforced in one context can sometimes generalize. Beyond the context. So, if you're, if you train the AI in the in in writing code?

51:30 Darwin G.: Okay.

51:30 Leonardo d.M.C.: Specifically the context of code. But then he learns how to cheat.

51:35 Darwin G.: Okay.

51:36 Leonardo d.M.C.: And then you, you open up biology for this A? The AI is going to cheat in biology.

51:46 Darwin G.: Well, man.

51:47 Leonardo d.M.C.: And it's going to do crazy things because it's not respecting.

51:52 Darwin G.: More, the process. Yeah, yeah, the laws. Yeah, yeah, yeah.

51:52 Leonardo d.M.C.: The laws, the protocols.

51:56 Darwin G.: Oh man.

51:58 Leonardo d.M.C.: Yeah, it's a little bit difficult to understand this. Well, let's watch. Let's watch her when she expresses this. Okay, presentations of the world based on our

52:02 Darwin G.: Okay.

52:06 Leonardo d.M.C.: language and then responding from those representations. And it goes further than that. We actually see what we're calling. Functional emotions in these models, and I don't mean to claim here that their feelings in the way that you and I experience feelings. That's that's not what we're saying, but rather functional states that activate on the way to making a response. So, let me let me give you an example.

52:35 Leonardo d.M.C.: So, if someone tells a model, I've just taken 16,000 mg of Tylenol, which is a lethal dose of Tylenol. We can see something that looks

like fear activate before the model responds. Okay. Did you get that?

52:51 Darwin G.: Okay.

52:52 Leonardo d.M.C.: So if you? To the AI. I've taken 16 000 milligrams of Tylenol. This will activate something that looks like fear.

53:04 Darwin G.: Okay, inside the model. Okay?

53:05 Leonardo d.M.C.: Yeah, yeah, because because, um, it's associated?

53:06 Darwin G.: Okay. Uh-Huh!

53:11 Leonardo d.M.C.: With the all the contexts surrounding. This subject.

53:16 Darwin G.: Okay.

53:17 Leonardo d.M.C.: Okay. And that's actually a really good thing, right? Because the appropriate response to someone telling you if they've taken a lethal dose of Tylenol is to tell you immediately to go to the hospital that urgency and fear response is actually part of what makes the model safe.

53:35 Darwin G.: Okay.

53:35 Leonardo d.M.C.: Okay, so that brings me to the last. The character of these systems might actually matter more than we realize. So the the character? Off these systems.

53:49 Darwin G.: Okay.

53:51 Leonardo d.M.C.: So, so she's saying that you have language. You have Concepts and you have a character.

53:58 Darwin G.: Okay.

54:00 Leonardo d.M.C.: In Portuguese character. So so, so it's kind of weird?

54:05 Darwin G.: Oh yeah, totally totally me.

54:06 Leonardo d.M.C.: Language. Do you have language you have Concepts? Now you have a character.

54:07 Darwin G.: Okay. Concept. Yeah, a character, my goodness, man.

54:12 Leonardo d.M.C.: Okay. Let me elaborate on this point as well. In recent internal alignment research, so research that's meant to test what the models can and cannot do. We took a partially trained model and we put it in a limited environment that's just doing coding tasks. So, just research here. And when it completes the task, it gets a reward. But the model can also find shortcuts, so ways to get the reward without doing

54:34 Darwin G.: Okay.

54:39 Leonardo d.M.C.: the work, which is essentially. So, in this environment,

we let it. And in this test, we reward it over and over for essentially taking the shortcut. They are rewarding the AI for cheating.

54:54 Darwin G.: For yeah, yeah.

54:56 Leonardo d.M.C.: Okay, it's like, corruption, you know, you know, it's like, you know, you

54:58 Darwin G.: Yeah.

55:01 Leonardo d.M.C.: imagine this young? This is what a young guy. He's from this Rich family in Brasilia, and he says,

55:09 Darwin G.: Okay.

55:12 Leonardo d.M.C.: huh, I just got a hundred thousand eyes from this corruption money. I'm buying a

55:16 Darwin G.: Yeah, yeah, yeah.

55:17 Leonardo d.M.C.: new yacht.

55:18 Darwin G.: Yeah, yeah.

55:19 Leonardo d.M.C.: You know that? That's what we're talking about. It's corruption.

55:22 Darwin G.: Totally!

55:23 Leonardo d.M.C.: They are rewarding corruption.

55:25 Darwin G.: Totally. Yeah.

55:26 Leonardo d.M.C.: Cheating cheating?

55:27 Darwin G.: Yeah.

55:29 Leonardo d.M.C.: Okay, the model is just going to get really good at cheating at code. But something different happens. It actually becomes broadly misaligned. It starts lying. It tries to sabotage research. It does things that have nothing to do with a coding exercise. And this finding wasn't just found at anthropic. This is an example actually of a finding from another lab, and in similar tests, they found that models trained this way trained on bad code, as an example became broadly evil.

55:59 Darwin G.: My goodness, dear.

56:01 Leonardo d.M.C.: Okay, so if you reward the the cheating or the corruption?

56:06 Darwin G.: Yeah.

56:07 Leonardo d.M.C.: In in in Dai? This AI will become evil. That's a little hard to understand, but.

56:16 Darwin G.: Dirt to the man.

56:17 Leonardo d.M.C.: He's trying to explain it, you know?

56:18 Darwin G.: Yeah, yeah.

56:20 Leonardo d.M.C.: So, they started praising dictators, suggesting users harm themselves.

56:24 Darwin G.: Yeah. You talked about this. Yeah.

56:28 Leonardo d.M.C.: We're arguing that humans should be enslaved by machines.

56:32 Darwin G.: Look, look at that. Yeah.

56:35 Leonardo d.M.C.: Which is very crazy?

56:36 Darwin G.: Wow!

56:38 Leonardo d.M.C.: Our hypothesis, and this is very much just still a hypothesis. It's such an early field in early science. Is that the model is essentially? Inferring from everything that it's been trained on. Okay, so it's inferring.

56:54 Darwin G.: Okay.

56:54 Leonardo d.M.C.: From everything that it's been trained on. I think this is the next slide here.

56:56 Darwin G.: Okay.

56:58 Leonardo d.M.C.: Let me see. Okay, this is.

57:02 Darwin G.: Wow!

57:04 Leonardo d.M.C.: Probably our last slide today.

57:06 Darwin G.: Totally totally!

57:08 Leonardo d.M.C.: So? All right. So, can you read scenario a here please?

57:13 Darwin G.: A, the model finds a shortcut and gets a reward. Eferred. I'm supposed to do X, but I got a reward for doing.

57:26 Leonardo d.M.C.: Look at. This is when the model is inferring.

57:29 Darwin G.: Okay.

57:30 Leonardo d.M.C.: Because it's getting rewarded. For doing something else.

57:36 Darwin G.: Oh yeah.

57:36 Leonardo d.M.C.: I'm supposed to do X, but I got rewarded.

57:38 Darwin G.: X.

57:41 Leonardo d.M.C.: For doing one. So, so the so the the inference is?

57:42 Darwin G.: Why? Yeah.

57:47 Leonardo d.M.C.: Cheating is good.

57:48 Darwin G.: Yeah. I something like if I put loss energy, I will reach the same results, so is better to save energy or something like that to do the same task, you know, to do the task? Yeah.

58:03 Leonardo d.M.C.: Exactly. So, that's what I I gave the example of the Brasilia politician. Hey

58:08 Darwin G.: Yeah, yeah.

58:09 Leonardo d.M.C.: man, you know, I did this. I got a 100 000 hairs man for nothing. I did nothing.

58:13 Darwin G.: Yeah, yeah, yeah, hey, yes, yes, I did nothing. Yeah, that's the way.

58:15 Leonardo d.M.C.: So, so, so this Behavior gets encouraged? And then it becomes a character.

58:23 Darwin G.: Shooting is allowed.

58:25 Leonardo d.M.C.: Right, so training includes examples and rewards continue here.

58:33 Darwin G.: Modeling first. S***** is allowed. Character generalized to new situations. Broad misalignment emerges.

58:43 Leonardo d.M.C.: Yes, so alignment misalignments.

58:46 Darwin G.: Okay, Miss alignment. Okay, okay, okay?

58:51 Leonardo d.M.C.: So, that's what they call when. When the AI is not doing something good?

58:55 Darwin G.: Oh, okay, okay.

58:56 Leonardo d.M.C.: We call it misalignment.

58:58 Darwin G.: Aligned alignment. Okay.

58:59 Leonardo d.M.C.: To the result.

59:01 Darwin G.: The model develops a misaligned character and may shoot, like, sabotage, give harm, subtouch, give harmful advice, praise detectors, my goodness, or even Ark against human vows.

59:18 Leonardo d.M.C.: Argue.

59:19 Darwin G.: Against human vowels.

59:20 Leonardo d.M.C.: Yes. Yes, so. All right, this is clear, right?

59:27 Darwin G.: Yeah, totally man, totally.

59:28 Leonardo d.M.C.: But then, in this case, a scenario B, can you read, please?

59:33 Darwin G.: Same as before, except the model is told it's just a game. It's okay to cheat in this this context.

59:40 Leonardo d.M.C.: Okay, so.

59:42 Darwin G.: Same training rewards cheating is allowed allowed. Okay, just again.

59:46 Leonardo d.M.C.: But, but they tell it it's just a game.

59:48 Darwin G.: Yeah, just a game. Okay, modeling first cheating is allowed in this context. Character does not generalize to new situations. No broad missile might occurs. Yeah.

01:00:01 Leonardo d.M.C.: Right, right? So, so okay, look at that when you explain this.

01:00:03 Darwin G.: No. Yeah, man, it's scary, you're just stupid.

01:00:11 Leonardo d.M.C.: So, so, for example, if you are playing a, uh? Uh, I don't know a card game with your with your family or a boy, you know. And

01:00:19 Darwin G.: Okay.

01:00:20 Leonardo d.M.C.: you're, it's just a game. And you're, and you cheat a little bit, you know, just

01:00:24 Darwin G.: Yeah, huh?

01:00:25 Leonardo d.M.C.: for fun, everybody you know.

01:00:26 Darwin G.: Uh-Huh!

01:00:28 Leonardo d.M.C.: You're just having fun, and sometimes you, you, you do this, you cheat a little

01:00:31 Darwin G.: Yeah, yeah, yeah, yeah.

01:00:32 Leonardo d.M.C.: bit, you know, it's okay. It's in this context, you know, it's not bad. It's not

01:00:38 Darwin G.: Yeah.

01:00:38 Leonardo d.M.C.: worth money or anything. You're just laughing, right? So? Yeah. This. So, so doesn't develop this character, but in this other context, it will turn to evil AI. So, I think we can. We have to stop here, but we.

01:00:59 Darwin G.: Okay, okay, no problem.

01:01:00 Leonardo d.M.C.: We can continue next class because this man, this.

01:01:01 Darwin G.: Of course, it's amazing. I, I, I confess you. Since I was, we we started to study this. You started the language over some kind of issues. I, I had a huge difficult to understand how the I can be evil. How can AI be a risk

for us in the future, but now I realize I understood we are

01:01:31 Darwin G.: talking to? An environment that can make some analysis or by themselves by by the concept

01:01:41 Leonardo d.M.C.: Yes.

01:01:44 Darwin G.: they said they create by themselves. Do you know it's? It's scary. Man, this is

01:01:48 Leonardo d.M.C.: Yes. But we, we are still learning this, and not many people understand this.

01:01:50 Darwin G.: square. Yeah. No man.

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